

<Observation>



<Instruction>

“Place the black bowl on the plate.”

TurboVLA inference



Vision Encoder

Text Encoder

Vision-
Language
Interaction

Action Head

Target FPGA scope

Robot Control

Action sequence

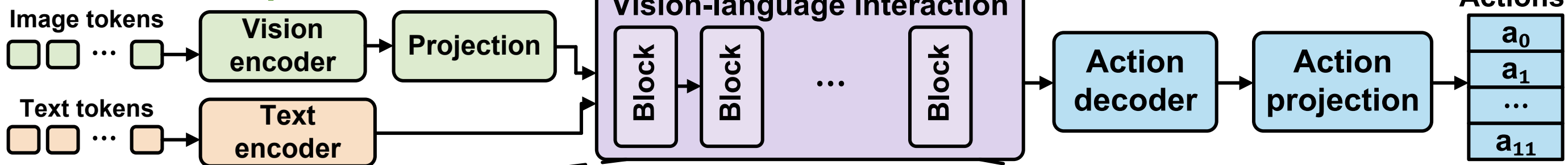
a_0 a_1 ... a_{11}
12 steps $\times$ 7 components

Controller

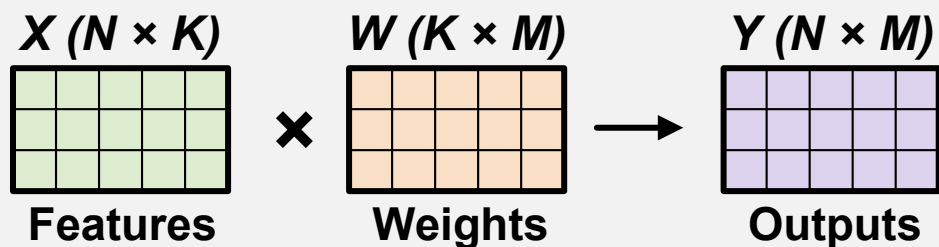


(a) TurboVLA-based robotic control

① Model computation



② Matrix computation



One output element

$$y = x_1 w_1 + x_2 w_2 + \dots + x_k w_k$$

Multiply and accumulate

Other operations:

Normalization

Activation

Layout

(b) Computation within one policy inference